

DISHA ACADEMY - ALL INDIA ENTRANCE TEST

JEE / NEET FOUNDATION PRACTICE PAPER

Time Allowed: 30 Minutes | Total Questions: 25 | Maximum Marks: 25

SECTION: PHYSICS

1. The dimensional formula for Planck's constant is identical to that of:

- (A) Linear momentum (B) Angular momentum
(C) Energy (D) Power

2. A body falls freely from rest under gravity. The ratio of distances travelled in 1st, 2nd and 3rd second is:

- (A) 1 : 2 : 3 (B) 1 : 3 : 5
(C) 1 : 4 : 9 (D) 1 : 1 : 1

3. Which of the following is a scalar quantity?

- (A) Electric field (B) Magnetic flux
(C) Acceleration (D) Torque

4. A particle moves in a circle of radius 20 cm with constant angular velocity 10 rad/s. Its linear speed is:

- (A) 1 m/s (B) 2 m/s
(C) 20 m/s (D) 0.5 m/s

5. The escape velocity from the surface of Earth does not depend on:

- (A) Mass of the earth (B) Radius of the earth
(C) Mass of the projectile (D) Gravitational constant

6. In simple harmonic motion, the acceleration of the particle is zero when:

- (A) Velocity is zero (B) Displacement is maximum
(C) Particle is at mean position (D) Potential energy is maximum

7. Two wires of same material have lengths in ratio 1:2 and diameters in ratio 2:1. The ratio of their resistances is:

- (A) 1 : 8 (B) 8 : 1
(C) 1 : 4 (D) 4 : 1

8. SI unit of magnetic flux is:

- (A) Tesla (B) Weber
(C) Henry (D) Gauss

SECTION: CHEMISTRY

9. The number of moles of solute present in 1 kg of solvent is called:

- (A) Molarity (B) Molality
(C) Normality (D) Mole fraction

10. Which of the following quantum numbers determines the shape of an orbital?

- (A) Principal (B) Azimuthal
(C) Magnetic (D) Spin

11. Which among the following elements has highest electron affinity?

- (A) Fluorine (B) Chlorine
(C) Bromine (D) Iodine

12. The hybridization of central carbon atom in methane (CH₄) is:

- (A) sp (B) sp²
(C) sp³ (D) sp³d

13. Which of the following is an amphoteric oxide?

- (A) Na₂O (B) SO₂
(C) Al₂O₃ (D) CaO

14. Oxidation state of Manganese in KMnO₄ is:

- (A) +2 (B) +4
(C) +6 (D) +7

15. The gas responsible for greenhouse effect in largest proportion is:

- (A) Methane (B) Carbon dioxide
(C) Ozone (D) Nitrous oxide

16. PH of a neutral aqueous solution at 25 degrees Celsius is:

- (A) 0 (B) 7
(C) 14 (D) 1

SECTION: MATHEMATICS

17. If matrix A is orthogonal, then determinant of A is equal to:

- (A) 0 (B) +1 or -1
(C) 2 (D) Any real number

18. The value of limit $x \rightarrow 0$ of $(\sin x / x)$ is equal to:

- (A) 0 (B) 1
(C) Infinity (D) Does not exist

19. Derivative of $e^{(2x)}$ with respect to x is:

- (A) $e^{(2x)}$ (B) $2 * e^{(2x)}$
(C) $2x * e^{(2x)}$ (D) $e^{(2x)} / 2$

20. The roots of equation $x^2 - 5x + 6 = 0$ are:

- (A) 2 and 3 (B) -2 and -3

(C) 1 and 6 (D) -1 and -6

21. The value of $\sin(90 - \theta)$ is equal to:

- (A) $\sin \theta$ (B) $\cos \theta$
(C) $-\cos \theta$ (D) $\tan \theta$

22. The sum of the first 10 natural numbers ($1 + 2 + \dots + 10$) is:

- (A) 45 (B) 50
(C) 55 (D) 60

23. The distance between points (0,0) and (3,4) is:

- (A) 3 (B) 4
(C) 5 (D) 7

24. The area of a circle with radius 7 cm (take $\pi = \frac{22}{7}$) is:

- (A) 154 cm^2 (B) 44 cm^2
(C) 88 cm^2 (D) 308 cm^2

25. If $\log_{10}(x) = 2$, then x is equal to:

- (A) 20 (B) 100
(C) 2 (D) 1000

ANSWER KEY

1. B 2. B 3. B 4. B 5. C 6. C 7. A 8. B
9. B 10. B 11. B 12. C 13. C 14. D 15. B 16. B
17. B 18. B 19. B 20. A 21. B 22. C 23. C 24. A 25. B